

Test Plan

Accounts Used

Admin: admin1 | Password: admin123

Instructor: inst1 | Password: inst1

Student: stu1, stu2 | Password: stu1, stu2

Login & Roles

1. Login with invalid credentials → Blocked
2. Login as role (stud/inst/admin) → Correct Dashboard Loads.
3. Wrong password 5 times → Change Password.
4. Auth DB only stores hash and not the password.
5. ERP DB contains no password fields.

Student Tests

1. Catalog shows all relevant course details.
2. Successfully register for or drop a course and section.
3. Cannot register in the same section twice.
4. Cannot register in full section.
5. Not possible to register for or drop a course after deadline
6. View grades for all registered courses.
7. Download transcript (CSV or PDF).
8. Can only view their own profile, timetable and notifications.

Instructor Tests

1. Sees only their own sections.
2. Enter different grade component scores.
3. Automatically compute final grade according to rule.
4. Can only view their own profile and class stats.
5. Download roster (CSV or PDF).
6. Can update or delete grade components.

Admin Tests

1. Creating a new user leads to both DB entries being created.
2. Can create a new course, section and assign instructors.
3. Toggle Maintenance ON leads to a banner being shown.
4. When ON, student/instructor actions are blocked.
5. When OFF, normal functionality is restored.
6. Can send out notifications to students.
7. Set registration and drop deadlines.
8. Can update or delete any details (except non-empty sections).
9. Possible to backup the database or restore older versions.

UI/UX

1. Buttons/labels are clear.
2. Errors are friendly.
3. Tables exist for a better understanding of data.
4. Searching and sorting implemented.
5. Colours, fonts, logos used to make the ERP appealing.

Test Summary

All tests were executed with the seeded sample data and accounts and all of them passed successfully.

Passed Tests

1. Login and role-based dashboards load correctly.
2. Student workflows all behave as expected.
3. Instructor workflows all behave as expected.
4. Admin workflows all behave as expected.
5. Password/auth separation is correctly implemented.
6. All edge and negative conditions were handled.
7. No data integrity issues observed.
8. Security checks and access rules are enforced.
9. UI/UX meets required criteria.
10. Performance is within acceptable ranges.
11. Maintenance mode is fully functional.
12. Backup/restore operates correctly.

Known Issues

Neither checked performance for large amounts of data or any specific long actions, nor implemented error handling for the same. Hence cannot be sure of no crashes or hangs.